

Notes: Kelly & Ljungqvist (RFS, 2012)

Testing Asymmetric-Information Asset Pricing Models

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1 Big picture

- **Question.** Do changes in information asymmetry affect equilibrium stock prices, investor demand, and expected returns in the way rational-expectations asset pricing models predict?
- **Shock.** The paper uses exogenous losses of public analyst signals caused by brokerage firms closing research departments.
- **Treatment.** Forty-three brokerage closures from Q1 2000 to Q1 2008 generate 4,429 coverage terminations for 2,180 U.S. stocks.
- **Mechanism.** A coverage termination removes a public signal. If institutions can replicate the signal privately but retail investors cannot, information asymmetry rises.
- **Main result.** Prices and retail demand fall, institutional demand rises, and liquidity-risk exposure increases.

2 Incremental contribution of the paper

- The paper provides a direct causal test of asymmetric-information asset pricing using broker closures rather than endogenous information-asymmetry proxies.
- It jointly tests price effects, investor-demand effects, comparative statics, and liquidity-risk implications.
- It distinguishes loss of public signals from bad news about firm fundamentals by using closure-induced terminations and filtering merger events.
- It connects information asymmetry to expected returns through systematic liquidity-risk exposure.

3 Data construction and measurement

The sample contains 43 brokerage closures and 4,429 coverage terminations for 2,180 unique firms. Coverage histories come from Reuters Estimates, I/B/E/S stop files, and Investext termination notices; announcements come from Factiva. The paper filters out noncommon stocks, firms without CRSP data, and firms near delisting. Merger-related observations are kept only when two pre-merger signals become one post-merger signal.

Interpretation. The construction is designed to isolate broker-level shocks to information production rather than analyst decisions to drop unattractive stocks.

3.1 Outcomes

Price effects are cumulative abnormal returns from market-model and Fama–French three-factor benchmarks. Information-asymmetry outcomes include bid-ask spreads, Amihud illiquidity, zero/missing return days, earnings-announcement volatility, and earnings surprises. Demand is measured with 13F institutional holdings.

Interpretation. The paper observes the information environment, the price response, and the ownership response.

4 Model and empirical specifications

4.1 Termination predictions

$$\Delta E[P_m] < 0, \quad \Delta E[InstitutionalDemand_m] > 0.$$

Interpretation. A public signal becomes private or unavailable to retail investors. Retail investors lower demand and the stock price falls until institutions absorb shares.

4.2 Initiation prediction

$$\Delta E[P_m] > 0, \quad \Delta E[InstitutionalDemand_m] < 0.$$

Interpretation. Coverage initiations are a reverse test: adding a public signal should raise prices and increase retail willingness to hold the stock.

4.3 Difference-in-differences

$$DiD_i = (post_i - pre_i) - (post_i^{control} - pre_i^{control}).$$

Interpretation. Controls remove secular trends in liquidity and information-asymmetry proxies.

4.4 Event-study CARs

$$CAR_i(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} abnormal\ return_{it}.$$

Interpretation. Negative CARs around closure announcements measure the price effect of losing public information.

4.5 Liquidity-risk test

The expected-return effect depends on the change in the Acharya–Pedersen total risk loading:

$$\Delta(\beta_1 + \beta_2 - \beta_3 - \beta_4).$$

Interpretation. If information asymmetry is priced through liquidity risk, terminated stocks should load more heavily on this priced combination after the shock.

5 Identification strategy

- **Exogenous variation.** Brokerage closures are broker-level events caused by the economics of research production, not stock-level information.
- **Sample filters.** Stand-alone closures are included. Merger-related observations are kept only when the stock loses one overlapping signal, not when the surviving broker chooses to drop coverage.
- **First-stage evidence.** Table 2 shows that information-asymmetry proxies rise after terminations.

- **No-bad-news evidence.** Figure 3 shows no pre-trend, and tests with imminent delistings and 9/11 terminations help rule out pure bad-news interpretations.
- **Comparative statics.** Effects are larger for retail brokers, local analysts, and rated analysts, consistent with signal-value predictions.
- **Main limitations.** Some lost signals may disappear entirely rather than become private; 13F measures only institutions; and event clustering requires bootstrap inference.

6 Tables and figures

6.1 Figure 1: Breakdown of brokerage closures and coverage terminations; Table 1: Closure-induced coverage terminations—Summary statistics

Read: Figure 1 shows the timing of closures and stock-level terminations. Table 1 shows treated firms are larger, more volatile, more liquid, and more covered than typical CRSP firms.

Interpretation. The shocks are broker-level and broadly spread over time; treated stocks are visible firms rather than obscure firms naturally losing attention.

Economic message. The sample is suitable for testing how losing a public signal affects actively traded stocks.

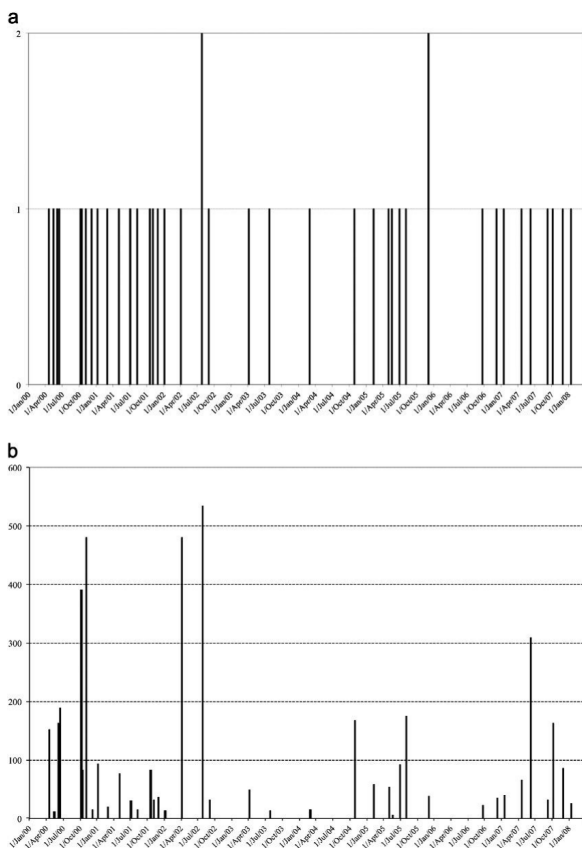


Figure 1
Breakdown of brokerage closures and coverage terminations

(a) Broker closures and terminations

Table 1
Closure-induced coverage terminations: Summary statistics

	Terminations sample	CRSP universe in 2004	Universe
Equity market value (\$ m)			
Mean	7,146.7	2,943.3	
Median	1,017.8	319.6	
Range	3.5–512,833	0.7–385,883	2.9
Monthly stock turnover			
Mean	0.20	0.18	
Median	0.12	0.09	
Range	0–3.0	0–17.9	
Daily return volatility (annualized %)			
Mean	52.1	45.1	
Median	60.0	39.1	
Range	11.6–214.3	8.1–277.4	8.
Number of brokers covering the stock			
Mean	10.6		
Median	9		
Range	1–40		
Number of unique firms	2,180	5,298	

The sample consists of 4,429 coverage terminations for 2,180 unique firms between Q1, 2000, and 1/2005. This table reports summary statistics for the market value of equity, share turnover (monthly volume of shares outstanding), daily return volatility, and the extent of coverage for each stock in the termination sample, the CRSP universe (share codes 10 and 11), and the universe of U.S. stocks with analyst coverage in the CRSP database. For each firm in the terminations sample, we calculate equity value and turnover for the month prior to the first termination date. For the universes of CRSP stocks and covered stocks, these variables are calculated for December 2004 (the midpoint of our sample period). Annualized volatility for terminations is the deviation of daily log returns in the twelve-month period that ends one month prior to a termination time. For the universes of CRSP stocks and covered stocks, volatilities are the annualized daily standard deviation of log returns for firms in these samples during calendar year 2004. (In each column, a firm is omitted from this calculation if it has fewer than 200 days of nonmissing returns.) The number of brokers who cover a stock in the terminations sample for the month prior to the termination is taken from the 13F SEC Form 13F summary data. (See

(b) Summary statistics

6.2 Figure 2: Average daily bid-ask spread, all CRSP firms, 2000-Q1 2008; Table 2: The effect of coverage terminations on information asymmetry

Read: Figure 2 shows secular liquidity trends. Table 2 shows treated firms experience increases in spreads, Amihud illiquidity, and earnings-announcement uncertainty relative to matched controls.

Interpretation. The termination changes the information environment, which is necessary for the asset-pricing test.

Economic message. The first-stage result validates the treatment as an information-asymmetry shock.

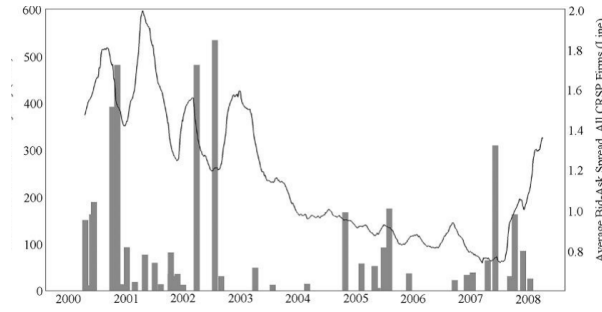


Figure 2
Average daily bid-ask spread, all CRSP firms, 2000-Q1 2008

The figure illustrates the extent to which proxies for information asymmetry have been subject to secular trends and swings over our sample period in the market as a whole. The figure shows trailing three-month average daily bid-ask spreads computed as $100 * (\text{ask} - \text{bid}) / [(\text{ask} + \text{bid}) / 2]$, averaged across the universe of firms in CRSP. Each day in our sample period. Similar secular trends and swings (available upon request) are observed in other proxies for information asymmetry, such as the log Amihud illiquidity measure and the number of days with zero or missing returns in CRSP. The quarterly number of closure-induced coverage terminations over the sample period is indicated by the grey vertical lines.

(a) Market-wide bid-ask spreads

Table 2
The effect of coverage terminations on information asymmetry

	Terminations		Matched controls		Mean DiD	p-value DiD = 0	<i>Econom magnitu. (percent change)</i>
	Before	After	Before	After			
Panel A: Bid-ask spreads							
3-mo. window	1.126	1.219	1.089	1.162	0.020	.011	1.8
6-mo. window	1.120	1.201	1.086	1.143	0.024	.010	2.1
Panel B: Amihud illiquidity measure (AIM)							
3-mo. window	0.104	0.147	0.099	0.127	0.015	<.001	14.6
6-mo. window	0.094	0.145	0.089	0.124	0.017	.001	18.0
Panel C: Missing and zero-return days (in % of days in the relevant window)							
3-mo. window	3.092	3.129	2.962	2.887	0.112	.080	3.6
6-mo. window	3.039	2.879	2.925	2.678	0.087	.123	2.9
Panel D: Earnings announcements							
Volatility (% p.a.)	37.82	38.80	33.24	28.81	5.418	<.001	14.3
Earnings surprise	0.637	0.744	0.647	0.667	0.088	<.001	13.8

The table reports cross-sectional means of proxies for information asymmetry before and after a termination event. For each sample termination, a control group is formed by selecting stocks with the same Daniel et al. (1997) size and book-to-market quintile assignment in the month of June prior to a termination, subject to the conditions that control firms 1) were covered by one or more analysts in the 12 months before the event; and 2) were not themselves subject to a coverage termination in the quarter before and after the event. When more than five matches exist, we choose the five stocks that are closest to the sample stock in terms of the relevant pre-event information asymmetry measure. (We lose 158 observations that involve sample stocks without viable controls and varying numbers of observations due to missing data necessary for calculating a particular measure. The number of observations ranges from about 4,100 to 4,300.) For each sample stock i , we construct a difference-in-difference test as $\text{DiD} = (\text{post}_i - \text{pre}_i) - (\text{post} - \text{pre})_{\text{Control Group}}$. The table reports the cross-sectional mean of the DiD statistic along with the mean percentage change in the DiD (mean before - 1). The latter illustrates the economic magnitude of the change in information asymmetry. We test the null hypothesis that information asymmetry is unchanged around a coverage termination using bootstrapped p -values. These adjust for potential cross-sectional dependence that is due to overlapping estimation windows caused by time clustering as multiple stocks are terminated in each brokerage firm closure. They are calculated by using a block bootstrap with a block length of 100, the approximate average number of terminations per brokerage closure event. The first three proxies for information asymmetry are computed over three- and six-month windows, which end ten days prior to the termination announcement or start ten days after the announcement date. Panel A reports changes in bid-ask spreads. Spreads are computed as $100 * (\text{ask} - \text{bid}) / [(\text{ask} + \text{bid}) / 2]$ using daily closing bid and ask data from CRSP. Panel B reports changes in the log Amihud illiquidity measure. This is defined as the natural log of one plus the ratio of the absolute stock return to the dollar trading volume and scaled by 10^6 (see Amihud 2002, p. 43). Panel C reports changes in the number of days with zero or missing returns in CRSP (using missing return codes -66, -77, -88, and -99). In Panel D, we report the effects of terminations on quarterly earnings announcements. The first measure is the annualized daily return volatility in a three-day window around earnings announcements for earnings announcements that occur in a one-year window before or after the drop. The second measure is the mean absolute value of quarterly earnings surprises in a one-year window before or after the drop. A surprise is defined as the absolute value of actual quarterly earnings minus the latest I/B/E/S consensus estimate before the earnings announcement, scaled by book value of equity per share, and multiplied by 100 for exposition.

(b) Information-asymmetry proxies

6.3 Figure 3: Daily abnormal returns around closure-induced coverage terminations; Table 3: Changes in price around coverage terminations

Read: Figure 3 shows no pre-trend, a sharp announcement drop, and no reversal. Table 3 reports negative CARs across event windows and benchmarks.

Interpretation. Prices fall when public information disappears, even though the event is not firm-specific bad news.

Economic message. Information asymmetry directly affects equilibrium prices.

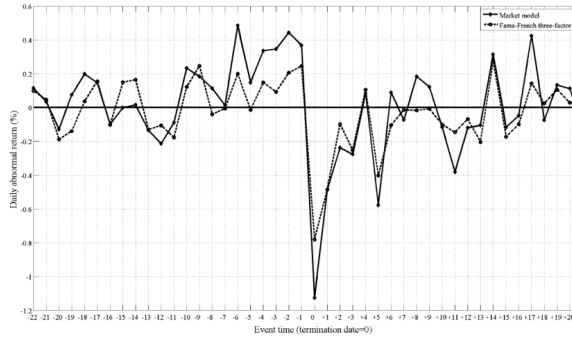


Figure 3
Daily abnormal returns around closure-induced coverage terminations
The terminations sample consists of 4,429 coverage terminations for 2,180 unique firms between Q1, 2000 and Q1, 2008. We compute abnormal returns for the forty-five trading days beginning on trading day -22 and ending on trading day +22, relative to the day a brokerage firm closure was announced (day 0). We use two alternative

(a) Abnormal returns

Table 3
Changes in price around coverage terminations

CAR calculated over the period . . .	No. of obs.	Market model	Fama-F 3-factor
Panel A: Closure-induced terminations			
Standard diff-in-diff approach:			
Close on day before termination to close on termination day	[-1, 0]	4,414 -1.12 ***	-0.78 *
Close on day before termination to close on day +1	[-1, +1]	4,414 -1.61 ***	-1.26 *
Close on day before termination to close on day +3	[-1, +3]	4,414 -2.12 ***	-1.61 *
Close on day before termination to close on day +5	[-1, +5]	4,414 -2.61 ***	-1.91 *
Close on day +5 after termination to close on day +22	[+5, +22]	4,414 0.19	-0.39
Portfolio strategy correcting for time clustering:			
Close on day before termination to close on termination day	[-1, 0]	43 -1.03 ***	-0.60
Close on day before termination to close on day +1	[-1, +1]	43 -1.33 ***	-0.86
Close on day before termination to close on day +3	[-1, +3]	43 -1.57 ***	-1.08
Close on day before termination to close on day +5	[-1, +5]	43 -1.85 ***	-1.26
Close on day +5 after termination to close on day +22	[+5, +22]	43 0.06	-0.78
Panel B: Terminations due to 9/11			
Standard diff-in-diff approach:			
Close on day before termination to close on termination day	[-1, 0]	356 -0.87 ***	-0.77 *
Close on day before termination to close on day +1	[-1, +1]	356 -0.90 ***	-0.65 *
Close on day before termination to close on day +3	[-1, +3]	356 -0.76 ***	-0.44
Close on day before termination to close on day +5	[-1, +5]	356 -1.08 ***	-0.62
Panel C: Imminent delistings			
Standard diff-in-diff approach:			
Close on day before termination to close on termination day	[-1, 0]	51 0.53	0.76
Close on day before termination to close on day +1	[-1, +1]	51 -0.32	-0.04
Close on day before termination to close on day +3	[-1, +3]	51 0.57	0.71
Close on day before termination to close on day +5	[-1, +5]	51 0.59	0.56

We compute cumulative abnormal returns over various windows around coverage terminations using benchmarks: the market model and the Fama-French three-factor model. Factor loadings are estimated in a year pre-event window ending eleven days before the termination date, and cumulative abnormal returns in the event window are calculated using this estimated model as a benchmark. (Results are nearly identical to those using a Carhart (1997) momentum factor in the Fama-French model.) We report these cumulative abnormal return metrics for the closures sample (Panel A); a separate sample of terminations at two brokerage firms in the World Trade Center that were devastated in the 9/11 attacks (Panel B); and a separate sample of coverage-induced terminations of stocks facing imminent delisting, for which a termination should be of little consequence (Panel C). To correct for after-hours announcements, we use time stamps to determine the first trading day after the announcement. Abnormal returns are reported in percent. In our specifications, they are averaged across all 4,414 sample stocks for which returns can be computed. (The sample is short of 4,429 because we require a minimum of fifty trading days of pre-event stock prices to estimate model parameters.) We report significance levels using a block bootstrap to control for dependence between events; we also report a standard event-study Patell *t*-test. These two test statistics are separated by “/.” In Panel A, we use a block length of 100 for the bootstrap, corresponding to the approximate number of terminations in the brokerage-closure event. In Panels B and C, given the much smaller sample sizes, we use block lengths of 5 and 5, respectively. In addition, to account for potential cross-sectional dependence caused by time clustering,

(b) Price effects

6.4 Table 4: Cross-stock spillover effects; Figure 4: Daily abnormal returns around industry-level coverage terminations

Read: Unaffected stocks in affected industries also fall. The spillovers are smaller than own-stock effects but significant.

Interpretation. Analyst research contains industry-level information, not only firm-specific information.

Economic message. Information asymmetry can spill across related assets.

Table 6
Changes in institutional holdings around coverage terminations

	No. of obs.	Terminations		Matched controls		Mean DID	p-value DID = 0	E_m $\sigma(\epsilon)$
		Before drop	After drop	Before drop	After Drop			
Panel A: Closure-induced terminations (in %)								
Standard diff-in-diff approach:	4,264	61.2	62.1	62.0	62.0	0.9	.043	
Portfolio strategy correcting for time clustering:	43	59.4	60.2	62.0	62.0	0.8	.034	
Panel B: Terminations due to 9/11 (in %)								
Standard diff-in-diff approach:	351	37.8	39.5	33.1	33.5	1.3	.294	
Panel C: Imminent delistings (in %)								
Standard diff-in-diff approach:	17	45.4	15.0	51.6	51.9	-30.7	.007	-

The table reports the quarterly change in institutional investors' holdings of stocks that experience coverage terminations. We report the mean fraction of total stock out that is held in aggregate by institutional investors filing 13f reports in the quarter before and the quarter after a termination. We then calculate a difference-in-difference $DID = (post_t - pre_t) - (DID^{control} - DID^{treatment})$, i.e., the difference between the pre- and posttermination change for sample stock i less the average change for control. We also report percentage changes ($\sigma(DID) / mean(DID) - 1$). Control groups are formed as described in Table 2. We report these statistics for the closures sample (Panel A), a sample of terminations at two brokerage firms located in the World Trade Center that were devastated in the 9/11 attacks (Panel B), and a separate sample of closure-induced term of stocks facing imminent delisting, for which a termination should be of little consequence (Panel C). The 13f data are taken from Thomson Reuters' CDA/Spectrum database. A, we lose 158 observations that involve sample stocks that have no viable controls and seven observations with missing 13f data. In Panel B, we similarly lose four observations valid matches and one observation without 13f data. In Panel C, we lose five observations without valid matches. In addition, because these are stocks that are about to be delisted, we lose thirty observations for which no 13f data are available in the quarter after the termination. We report both standard diff-in-diff estimates, treating each observation as independent using a block bootstrap to control for dependence among events, and (in Panel A) a portfolio strategy that buys all stocks that are terminated on a given closure day. In the latter estimate of the average change in institutional holdings in the portfolio strategy is the time-series average of the forty-three brokers' average change in institutional holdings, in with a Fama-MacBeth regression; significance levels are computed from the standard deviation of the forty-three averages. In Panel A, we use a block length of 100 for the b corresponding to the approximate number of terminations per brokerage closure event. In Panel B, given the much smaller sample sizes, we use a block length of 10. In Panel C, n have only seventeen observations due to the fact that most companies in this sample are delisted before the next 13f filing date, we compute a simple t -test.

Institutional ownership changes

Table 7
Cross-sectional determinants of changes in price and in institutional demand

		Dependent variable:					
		CAR [-1,0] in %			DiD change in 13f holdings		
		(1)	(2)	(3)	(4)	(5)	(6)
μ	13f Holdings in quarter before drop	1.250*** 0.394	2.072*** 0.554	-0.076 0.492	-0.060*** 0.009	-0.061*** 0.011	-0.062* 0.013
$\bar{X}$	Mean daily log turnover	-0.635*** 0.108	-1.047*** 0.165	-0.112 0.141	0.004* 0.002	0.007* 0.003	0.001 0.003
η^{-1}	SD of log turnover	-1.235*** 0.275	-1.443*** 0.423	-1.045** 0.343	-0.007 0.005	0.004 0.007	-0.019 0.019
ρ^{-1}	SD of earnings per share	-0.327 [†] 0.177	-0.519 [†] 0.289	-0.081 0.183	0.035*** 0.009	0.041*** 0.011	0.029* 0.013
γ^{-1}	Analyst forecast dispersion	-2.237* 0.882	-3.769** 1.372	-0.029 0.906	-0.009 0.039	0.007 0.054	-0.010 0.059
γ^{-1}	Analyst forecast dispersion ²	0.869* 0.396	1.417* 0.577	-0.023 0.473	-0.088 0.076	-0.181 [†] 0.110	-0.020 0.113
γ^{-1}	= 1 If analyst is local	-0.349* 0.160	-0.551* 0.258	-0.008 0.168	-0.003 0.004	-0.005 0.005	0.001 0.005
γ^{-1}	= 1 If analyst is rated	-0.205 0.235	0.063 0.267	-0.743* 0.308	0.006 0.005	0.002 0.006	0.027 0.017
γ^{-1}	Log experience of covering analyst	-0.375*** 0.103	-0.616*** 0.156	-0.059 0.106	0.000 0.002	0.003 0.002	-0.006 0.004
	= 1 If coincides w/ neg. earnings surprise	-1.605 1.081	-3.916* 1.582	1.110 1.153	0.006 0.016	-0.012 0.022	0.028 0.024
	Log no. other brokers covering the stock	-0.010 0.116	-0.052 0.174	0.055 0.138	0.000 0.002	0.000 0.003	-0.001 0.004
	Adjusted R^2	13.3%	16.0%	5.7%	4.0%	4.6%	4.0%
	Wald test: all coeff. = 0	75.5***	83.3***	21.9*	75.7***	62.4***	36.6***
	No. of observations	4,321	2,446	1,875	4,162	2,342	1,821

We test the model's comparative statics by regressing proxies for changes in price and institutional demand around coverage terminations on proxies for μ (the fraction of informed investors); $\bar{X}$ (mean aggregate supply η^{-1} (aggregate supply uncertainty); ρ^{-1} (payoff uncertainty); and γ^{-1} (signal noise). The proxy for price changes is the market-adjusted CAR from the day before the announcement of a brokerage-firm closure to the end of the announcement day. The proxy for informed-demand changes is the difference-in-difference (DiD) change in the fraction of the company's stock held by institutional investors from the quarter before to the quarter after a termination, net of the mean contemporaneous change in institutional holdings in matched control firms (see Table 6 for further details). The independent covariates are defined as follows. The proxy for μ is the fraction of the company's stock that is held by institutional investors as of the quarter-end prior to the termination. Mean and variance of aggregate supply are based on the first two moments of the distribution of log daily turnover which is estimated over the six months that end one month prior to the termination. Our proxy for payoff variance is the standard deviation of quarterly earnings per share, using up to twenty quarters of data prior to the termination. Our proxies for signal noise include the level and square of analyst forecast dispersion (defined as the time-series mean of the standard deviation of analyst earnings per share forecasts in the year prior to the termination); whether the analyst is local to the company suffering a termination (defined as being local within a 200-mile radius of its headquarters); whether the analyst is an "all-star" or a *Wall Street Journal* "mas stock picker"; and for the analyst's experience (the log number of quarters since the analyst first appeared in I/B/E/S databases or, if missing, when he or she obtained their first license according to the Financial Industry Regulatory Authority's "Broker Check" service). Where necessary, variables are winsorized to reduce noise and avoid outliers driving the results. We control for whether the termination coincides with a negative earnings surprise and for the number of remaining analysts that cover the stock (estimated from the I/B/E/S forecast summary file). All specifications include broker fixed effects. These remove observed and unobserved firm

Cross-sectional tests

6.7 Table 7: Cross-sectional determinants of changes in price and institutional demand

Read: Responses vary with institutional ownership, turnover, volatility, and signal precision proxies.

Interpretation. The paper tests not only an average effect but the model’s comparative statics.

Economic message. Effects are stronger where the lost signal should matter more.

6.8 Table 8: The effect of coverage terminations on liquidity risk exposure

Read: Total Acharya–Pedersen liquidity-risk loading rises after coverage terminations.

Interpretation. Information asymmetry affects expected returns through priced liquidity risk.

Economic message. The liquidity channel links information asymmetry to asset pricing.

Table 8
The effect of coverage terminations on liquidity risk exposure

	Terminations		Matched controls		Mean DiD	<i>p</i> -value
	Before drop	After drop	Before drop	After drop		
Panel A: 12-mo. window						
Net DiD: $\Delta(\beta_1 + \beta_2 - \beta_3 - \beta_4)$					0.168	
β_1 (stock return vs. market return)	1.177	1.143	1.122	1.090	-0.003	
β_2 (stock illiquidity vs. mkt illiquidity)	0.016	0.026	0.046	0.039	0.017	
β_3 (stock return vs. market illiquidity)	-0.032	-0.023	-0.026	-0.020	0.003	
β_4 (stock illiquidity vs. market return)	-0.112	-0.174	-0.233	-0.139	-0.156	
Panel B: 18-mo. window						
Net DiD: $\Delta(\beta_1 + \beta_2 - \beta_3 - \beta_4)$					0.191	
β_1 (stock return vs. market return)	1.176	1.162	1.114	1.128	-0.028	
β_2 (stock illiquidity vs. mkt illiquidity)	0.016	0.031	0.042	0.048	0.009	
β_3 (stock return vs. market illiquidity)	-0.027	-0.024	-0.024	-0.022	0.001	
β_4 (stock illiquidity vs. market return)	-0.146	-0.330	-0.219	-0.192	-0.211	
Panel C: 24-mo. window						
Net DiD: $\Delta(\beta_1 + \beta_2 - \beta_3 - \beta_4)$					0.164	
β_1 (stock return vs. market return)	1.167	1.174	1.101	1.139	-0.031	
β_2 (stock illiquidity vs. mkt illiquidity)	0.019	0.035	0.045	0.049	0.013	
β_3 (stock return vs. market illiquidity)	-0.027	-0.024	-0.023	-0.022	0.002	
β_4 (stock illiquidity vs. market return)	-0.165	-0.337	-0.267	-0.254	-0.184	

The table reports cross-sectional means of exposure to systematic liquidity risk before and after termination for sample stocks and their matched controls. Changes in liquidity risk are assessed within the framework of Acharya and Pedersen (2005), who propose an equilibrium model that describes how individual firms’ exposure to aggregate liquidity risk affects expected returns. The resulting pricing equation $E(r_{i,t}^* | r_{f,t}) = E(c_{i,t}) + \lambda(\beta_{1t} + \beta_{2t} - \beta_{3t} - \beta_{4t})$, is described in Section III.D. The four betas capture exposure to aggregate risks embodied by the comovement between stock returns and the market return (β_1); stock illiquidity and aggregate illiquidity (β_2); stock returns and aggregate illiquidity (β_3); and stock illiquidity and the market return (β_4). Increases in expected returns that are due to liquidity risk are associated with an increase in β_1 and decreases in β_3 and β_4 . Our empirical methodology closely follows Acharya and Pedersen (2005), except that we use a weekly frequency (Wednesday close to Wednesday close). For each stock, illiquidity is calculated as the adjusted Amihud measure, which is defined as $\min(0.25 + 0.30 \text{Illiq}_t^M, 30.00)$, where Illiq_t^M is the absolute stock return to the dollar trading volume (scaled by 10^6) for stock i averaged over the days in week t and Illiq_{t-1}^M is the ratio of the capitalizations of NYSE and AMEX stocks at the end of week $t-1$ to the value-weighted average illiquidity of NYSE and AMEX stocks each week. Betas are calculated by using innovations in stock illiquidity and aggregate illiquidity defined as residuals from AR(2) models because of the persistence in illiquidity levels. As in previous tables, for each sample termination, a control group is formed by selecting stocks with the same Daniel et al. (1995) and book-to-market benchmark assignment in the month of June prior to a termination, subject to the condition that control firms (1) were covered by one or more analysts in the three months before the event; and (2) did not themselves experience a coverage termination in the quarter before and after the event. When more than five matches exist, we choose the five stocks closest to the sample stock in terms of market beta. (Of the 158 observations that involve stocks without any viable controls.) Betas are winsorized to 2.5% in each direction to mitigate the effect of extreme outliers and reduce noise. We then calculate a difference-in-differences test for each sample stock i , $\text{DiD} = (\text{post}_i - \text{pre}_i) - (\text{post}_{\text{Control Group } i} - \text{pre}_{\text{Control Group } i})$, and report the cross-sectional mean. Betas are calculated by using weekly data over twelve-, eighteen-, and twenty-four-month windows ending two weeks prior to the termination announcement or start two weeks after the announcement date. The null hypothesis that exposure to systematic liquidity risk is unchanged around a coverage termination is tested using bootstrapped p -values. These adjust for potential cross-sectional dependence that is due to overlapping windows caused by time clustering as multiple stocks are terminated in each brokerage-firm closure. The bootstrap uses a block length of 100, corresponding to the approximate number of terminations per

Liquidity-risk loadings

6.9 Table 9: Changes in price around coverage initiations; Figure 5: Daily abnormal returns around coverage initiations

Read: Coverage initiations generate positive abnormal returns, providing a reverse test.

Interpretation. Adding a public signal reduces asymmetry and raises prices.

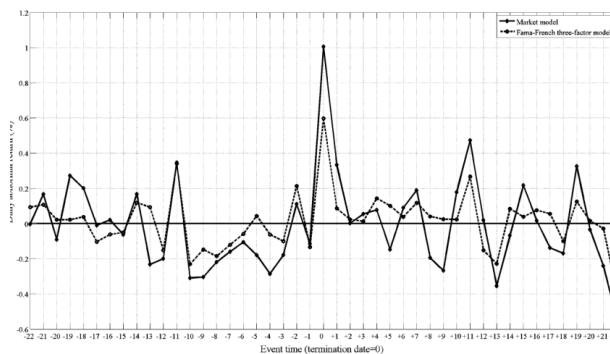
Economic message. Initiation evidence reinforces the causal interpretation.

Table 9
Changes in price around coverage reinitiations

Cumulative abnormal return (CAR) calculated over the period . . .	No. of obs.	Market model	Fama-Fr 3-factor
Close on day before reinitiation to close on reinitiation day	[-1, 0]	1,010	1.01 ***/*** 0.61 *
Close on day before reinitiation to close on day +1	[-1, +1]	1,010	1.34 ***/*** 0.70 *
Close on day before reinitiation to close on day +3	[-1, +3]	1,010	1.39 ***/*** 0.73 *
Close on day before reinitiation to close on day +5	[-1, +5]	1,010	1.30 ***/*** 0.95 *
Close on day +5 after reinitiation to close on day +22	[+5, +22]	1,010	-0.55 -0.03

We compute cumulative abnormal returns over various windows around coverage reinitiations by using benchmarks: the market model and the Fama-French three-factor model. Factor loadings are estimated in year pre-event window that ends eleven days before the reinitiation date, and cumulative abnormal returns the event window are calculated by using the estimated model as a benchmark. (Results are nearly identical we include a Carhart (1997) momentum factor in the Fama-French model.) Abnormal returns are reported in percentages. The reinitiations sample consists of 1,019 coverage reinitiations for 902 unique firms between 1997 and Q1, 2008. We lose nine observations because we require a minimum of fifty trading days of pre-stock prices in order to estimate model parameters. We report significance levels by using a block bootstrap length 100 in order to control for dependence among events; we also report a standard event-study Patel. These two test statistics are separated by "/." We use ***, **, and * to denote statistical significance at the 1%, and 5% levels (two-sided), respectively.

The remaining six mergers involve 1,220 stocks covered by the institut broker but not by the retail broker ahead of the merger. We identify reinit



(a) Initiation price effects

(b) Initiation abnormal returns

7 Interpretive synthesis

Information asymmetry changes prices, ownership, and liquidity risk. The core mechanism is that losing a public signal raises uncertainty for uninformed investors while leaving informed investors better positioned to absorb shares. The evidence is strongest because it combines exogenous events, first-stage changes in information asymmetry, price reactions, institutional demand changes, comparative statics, and a reverse initiation experiment.

8 Short summary

- Brokerage closures provide 4,429 exogenous coverage terminations.
- Information-asymmetry proxies rise after terminations.
- Prices fall and do not reverse.
- Institutions increase holdings while retail investors are inferred to sell.
- Effects are larger for more public and precise lost signals.
- Liquidity-risk loadings rise, linking information asymmetry to expected returns.
- Coverage initiations produce positive price effects.

9 Conclusion

Kelly and Ljungqvist show that asymmetric information is empirically important for asset pricing. Public analyst information affects equilibrium prices, ownership, and expected returns. The paper's main lesson is that information markets and liquidity risk are connected: when public information disappears, prices fall, informed investors hold more, and the stock becomes more exposed to liquidity risk.